State Arrays for Single Particle Tracking (SASPT)

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Alec Heckert, Liza Dahal, Robert Tjian, Xavier Darzacq (2022) Recovering mixtures of fast-diffusing states from short single-particle trajectories eLife https://doi.org/10.7554/eLife.70169

Theory

Definitions

Define: $\boldsymbol{Z}$ a matrix indicating the state of a particular trajectory. For N trajectories and K states.

$$Z \in \{0,1\}^{N \times K}$$

Define: $\boldsymbol{\tau}$ as a vector of probabilities of a trajectory being in a certain state. τ_k is the probability of a trajectory being in state k.

Define: $\boldsymbol{\theta}$ as a vector of parameters defining motion (ex: diffusion coefficient).

Aside on Bayesian statistics

Bayesian statistics is based on Bayes' rule

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)} = \frac{P(\theta, D)}{P(D)}$$

Posterior probability $P(\theta|D)$: Probability after data is collected (updated belief).

Likelihood $P(D|\theta)$: Probability of seeing the data given model parameters (explains the fit of a statistical model since likelihood explains the likelihood of the data being explained by θ).

Prior probability $P(\theta)$: Probability before data is collected (original belief).

Marginal probability $P(D)$: Probability of recording data D.

Variational Approach

We only observe trajectories $\boldsymbol{X}$ but wish to infer $\boldsymbol{Z}$ and $\boldsymbol{\tau}$. If we can estimate $p(\boldsymbol{Z}, \boldsymbol{\tau}|\boldsymbol{X})$, we can estimate $\boldsymbol{Z}$ and $\boldsymbol{\tau}$.

Problem: $p(\boldsymbol{Z}, \boldsymbol{\tau}|\boldsymbol{X})$ is intractable to calculate in practice.

$$p(\boldsymbol{Z}, \boldsymbol{\tau}|\boldsymbol{X}) = \frac{p(\boldsymbol{Z}, \boldsymbol{\tau}, \boldsymbol{X})}{p(\boldsymbol{X})}$$

Marginal probability $p(\boldsymbol{X}) = \sum_{\boldsymbol{Z}} \int_{\tau} p(\boldsymbol{Z}, \boldsymbol{\tau}, \boldsymbol{X}) d\boldsymbol{\tau}$ requires an integral over all state probabilities (i.e. this is impossible to calculate).

Solution: We can take a variational Bayesian approach whereby we define a calculable surrogate $q(\boldsymbol{Z}, \boldsymbol{\tau}) \approx p(\boldsymbol{Z}, \boldsymbol{\tau}|\boldsymbol{X})$.

Variational Approach – Derivation

How do we optimize $q(\boldsymbol{Z}, \boldsymbol{\tau})$?

We should choose q such that the KL-Divergence $D(q||p)$, a measure of distance between distributions, between q and p is minimized.

$$q = \underset{q}{\operatorname{argmin}}\, D(q||p)$$

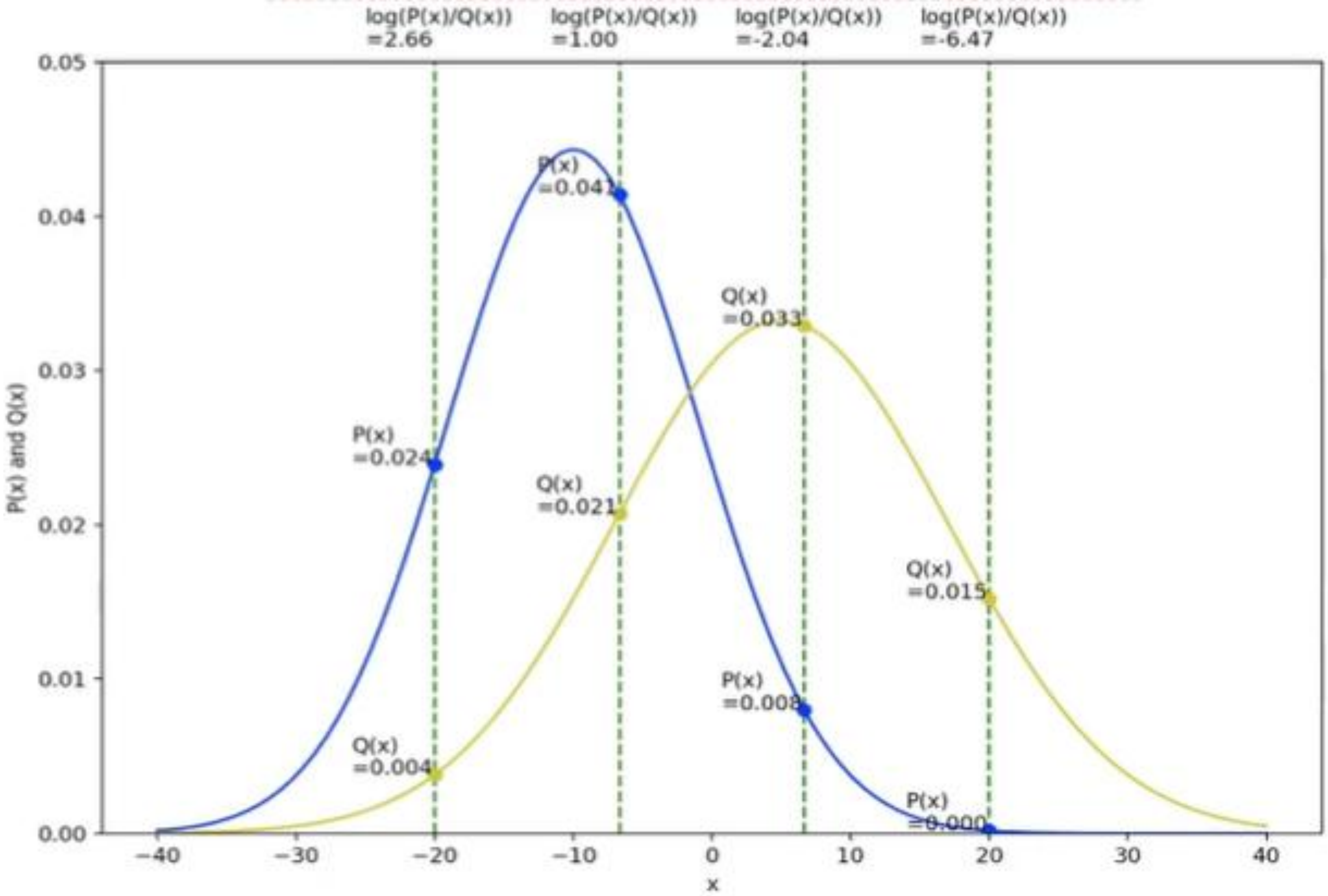
But… we still don't know what p is, so D is incalculable.

Aside on KL-Divergence

Definition: KL-Divergence measures how different two probability distributions are. It is defined as

$$D(p||q) = \mathbb{E}_{\boldsymbol{x}\sim p(\boldsymbol{x})}\left[\log\frac{p(\boldsymbol{x})}{q(\boldsymbol{x})}\right] = \int_{\boldsymbol{x}} p(\boldsymbol{x})\log\frac{p(\boldsymbol{x})}{q(\boldsymbol{x})}\,d\boldsymbol{x}$$

Intuition: $\log\frac{p(x)}{q(x)} > 0$ for $p(x) > q(x)$ and $\log\frac{p(x)}{q(x)} < 0$ for $p(x) < q(x)$. Weight each ratio by $p(x)$ since positive ratios have a large $p(x)$ and vice versa. Unweighted sums sum to 0.



log(P(x)/Q(x))
=2.66
log(P(x)/Q(x))
=1.00
log(P(x)/Q(x))
=-2.04
log(P(x)/Q(x))
=-6.47
P(x)
=0.041
Q(x)
=0.033
P(x)
=0.024
Q(x)
=0.021
Q(x)
=0.015
P(x)
=0.008
Q(x)
=0.004
P(x)
=0.000
0.05
0.04
0.03
0.02
0.01
0.00
P(x) and Q(x)
−40
−30
−20
−10
0
10
20
30
40
x

Variational Approach – Derivation

Let's convert the KL-Divergence into a calculable problem.

$$
\begin{aligned}
D(q||p) &= \sum_{\boldsymbol{Z}} \int_{\boldsymbol{\tau}} q(\boldsymbol{Z}, \boldsymbol{\tau}) \log \frac{q(\boldsymbol{Z}, \boldsymbol{\tau})}{p(\boldsymbol{Z}, \boldsymbol{\tau}|\boldsymbol{X})} d\boldsymbol{\tau} = \sum_{\boldsymbol{Z}} \int_{\boldsymbol{\tau}} q(\boldsymbol{Z}, \boldsymbol{\tau}) \log \frac{q(\boldsymbol{Z}, \boldsymbol{\tau}) p(\boldsymbol{X})}{p(\boldsymbol{Z}, \boldsymbol{\tau}, \boldsymbol{X})} d\boldsymbol{\tau} \\
&= \sum_{\boldsymbol{Z}} \int_{\boldsymbol{\tau}} q(\boldsymbol{Z}, \boldsymbol{\tau}) \log \frac{q(\boldsymbol{Z}, \boldsymbol{\tau})}{p(\boldsymbol{Z}, \boldsymbol{\tau}, \boldsymbol{X})} d\boldsymbol{\tau} + \sum_{\boldsymbol{Z}} \int_{\boldsymbol{\tau}} q(\boldsymbol{Z}, \boldsymbol{\tau}) \log p(\boldsymbol{X}) \, d\boldsymbol{\tau} \\
&= \mathbb{E}\left[\log \frac{q(\boldsymbol{Z}, \boldsymbol{\tau})}{p(\boldsymbol{Z}, \boldsymbol{\tau}, \boldsymbol{X})}\right] + \mathbb{E}[log\, p(\boldsymbol{X})] \\
&= -\sum_{\boldsymbol{Z}} \int_{\boldsymbol{\tau}} q(\boldsymbol{Z}, \boldsymbol{\tau}) \log \frac{p(\boldsymbol{Z}, \boldsymbol{\tau}, \boldsymbol{X})}{q(\boldsymbol{Z}, \boldsymbol{\tau})} d\boldsymbol{\tau} + \log p(\boldsymbol{X})
\end{aligned}
$$

Define $L[q]$ such that

$$D(q||p) = -L[q] + \log p(\boldsymbol{X})$$

$L[q]$ is called the variational lower bound or evidence lower bound (ELBO).

Variational Approach – Derivation

We now have a calculable value

$$L[q] = \sum_{\boldsymbol{Z}} \int_{\boldsymbol{\tau}} q(\boldsymbol{Z}, \boldsymbol{\tau}) \log \frac{p(\boldsymbol{Z}, \boldsymbol{\tau}, \boldsymbol{X})}{q(\boldsymbol{Z}, \boldsymbol{\tau})} \, d\boldsymbol{\tau}$$

Note that maximizing $L[q]$ minimizes $D(q||p)$ since $\log p(\boldsymbol{X})$ is a constant.

Now let's find closed form equations for all needed functions.

Variational Approach – Priors

Before we find closed forms, we need to define priors for $\boldsymbol{\tau}, \boldsymbol{Z}$, and $\boldsymbol{X}$.

$$
\begin{aligned}
\boldsymbol{\tau} &\sim \mathrm{Dir}\left(\frac{\alpha}{K}, \dots, \frac{\alpha}{K}\right) \\
Z_i \mid \boldsymbol{\tau} &\sim \mathrm{Multinomial}(\boldsymbol{\tau}, 1) \\
X_i \mid Z_{ik} = 1 &\sim p_{\boldsymbol{X}}(x \mid \theta_k)
\end{aligned}
$$

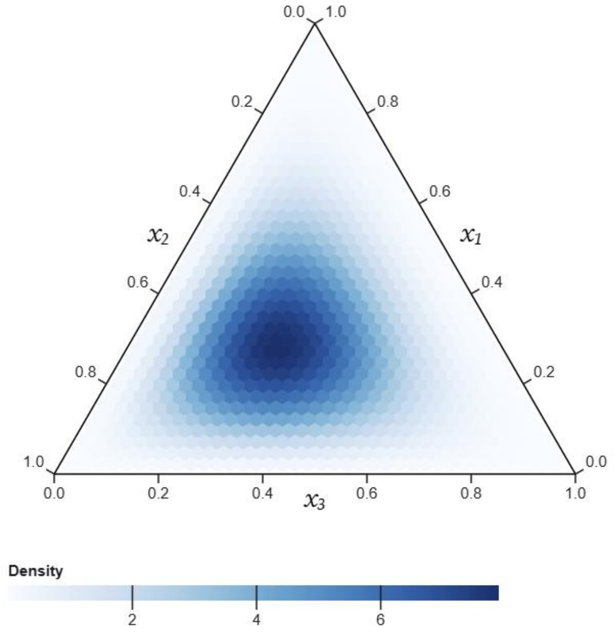
For Dirichlet distribution Dir and model of motion $p_{\boldsymbol{X}}(x \mid \theta_k)$ such as random Brownian motion.

Aside on Dirichlet distribution

The Dirichlet distribution models probability of probabilities.

EX: A student is seen wearing a red shirt 2 days, a blue shirt 3 days, and a green shirt 2 days of a given week. We use the Dirichlet distribution to model the probability vector that the student wears a certain color shirt the next day.

$$\mathrm{Dir}(3,4,3)$$



Definition:

$$\mathrm{Dir}(\boldsymbol{x}|\alpha_1, \dots, \alpha_k) = \frac{1}{B(\boldsymbol{\alpha})} \prod_{i=1}^{k} x_i^{\alpha_i - 1}$$

Variational Approach – Closed Forms

Let's start with $q(\boldsymbol{Z}, \boldsymbol{\tau}) = q(\boldsymbol{Z})q(\boldsymbol{\tau})$.

Through variational calculus (see Mean Field Approximation), it can be shown that

$$\log q(\boldsymbol{Z}) = \mathbb{E}_{\boldsymbol{\tau}\sim q(\boldsymbol{\tau})}[\log p(\boldsymbol{Z}, \boldsymbol{\tau}, \boldsymbol{X})] + C$$
$$\log q(\boldsymbol{\tau}) = \mathbb{E}_{\boldsymbol{Z}\sim q(\boldsymbol{Z})}[\log p(\boldsymbol{Z}, \boldsymbol{\tau}, \boldsymbol{X})] + C$$

Variational Approach – Closed Forms

We can solve for q once we have p.

$$\begin{aligned}\log p(\boldsymbol{Z}, \boldsymbol{\tau}, \boldsymbol{X}) &= \log p(\boldsymbol{Z}|\boldsymbol{\tau}) + \log p(\boldsymbol{\tau}) + \log p(\boldsymbol{X}|\boldsymbol{Z}) \\ &= \sum_{k=1}^{K} \sum_{i=1}^{N} Z_{ik} \log \tau_k + \sum_{k=1}^{K} \left(\frac{\alpha}{K} - 1\right) \log \tau_k + \sum_{k=1}^{K} \sum_{i=1}^{N} Z_{ik} \log A_{ik} + C\end{aligned}$$

Let's plug this into our equations for $q(\boldsymbol{\tau})$ and $q(\boldsymbol{Z})$ on the previous slide.

Variational Approach – Closed Forms

$$\log q(\boldsymbol{\tau}) = \mathbb{E}_{\mathbf{Z}\sim q(\mathbf{Z})}\left[\sum_{k=1}^{K}\sum_{i=1}^{N} Z_{ik} \log \tau_k + \sum_{k=1}^{K}\left(\frac{\alpha}{K}-1\right)\log \tau_k + \sum_{k=1}^{K}\sum_{i=1}^{N} Z_{ik}\log A_{ik}\right] + C$$
$$= \sum_{k=1}^{K}\left(\frac{\alpha}{K} - 1 + \sum_{i=1}^{N}\mathbb{E}_{\mathbf{Z}\sim q(\mathbf{Z})}[Z_{ik}]\right)\log \tau_k + C$$

Hence

$$q(\boldsymbol{\tau}) = Dir(n_1, \ldots, n_k)$$
$$n_k = \frac{\alpha}{K} + \sum_{i=1}^{N} L_i \mathbb{E}_{\mathbf{Z}\sim q(\mathbf{Z})}[Z_{ik}]$$

Note we added the L_i weight (number of jumps in trajectory i) to give longer trajectories more weight.

Variational Approach – Closed Forms

$$\log q(\boldsymbol{Z}) = \mathbb{E}_{\boldsymbol{\tau}\sim q(\boldsymbol{\tau})}\left[\sum_{k=1}^{K}\sum_{i=1}^{N} Z_{ik}\, log\, \tau_k + \sum_{k=1}^{K}\left(\frac{\alpha}{K} - 1\right) log\, \tau_k + \sum_{k=1}^{K}\sum_{i=1}^{N} Z_{ik}\, log\, A_{ik}\right] + C$$

$$= \sum_{k=1}^{K}\sum_{i=1}^{N}\left(\log A_{ik} + \mathbb{E}_{\boldsymbol{\tau}\sim q(\boldsymbol{\tau})}[\log \tau_k]\right) Z_{ik} + C$$

We know $\mathbb{E}_{\boldsymbol{\tau}\sim q(\boldsymbol{\tau})}[\log \tau_k] = \psi(n_k) - \psi(\sum n_i)$ for the digamma function ψ from properties of the Dirichlet distribution.

Hence

$$q(\boldsymbol{Z}) = \prod_{i=1}^{N}\prod_{k=1}^{K} r_{ik}^{Z_{ik}}$$

$$r_{ik} = \frac{A_{ik}\exp\psi(n_k)}{\sum_{j=1}^{K} A_{ij}\exp\psi(n_j)}$$

After normalizing for each trajectory.

State Array Algorithm

1. Initialize matrix $\boldsymbol{A}$ with $A_{ik} = p(X_i|\theta_k)$
2. Initialize matrix $\boldsymbol{r}$ with $r_{ik}^{<0>} = \frac{A_{ik}}{\sum_{j=1}^{K} A_{ij}}$
3. For iteration $t = 1,2,3, \ldots$
   a) $n_k^{<t>} = \frac{\alpha}{K} + \sum_{i=1}^{N} L_i r_{ik}^{<t-1>}$
   b) $r_{ik}^{<t>} = \frac{A_{ik} \exp \psi\left(n_k^{<t>}\right)}{\sum_{j=1}^{K} A_{ij} \exp \psi(n_j^{<t>})}$

*Convergence is guaranteed by Pythagorean theorem for Bergman divergence

Code

Initialization

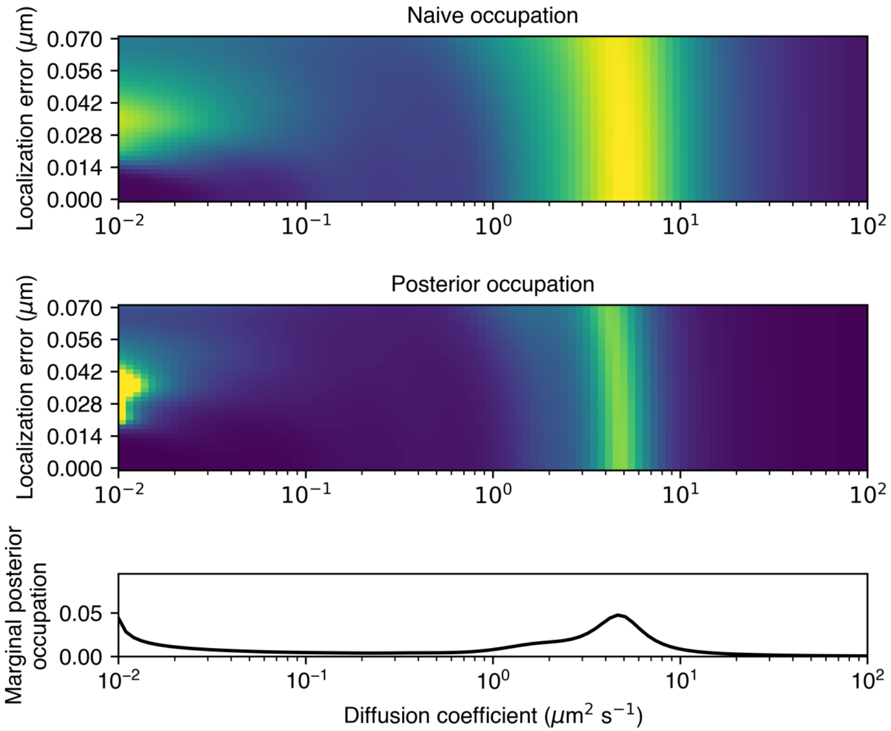
StateArray.from_detections(detections, **settings)

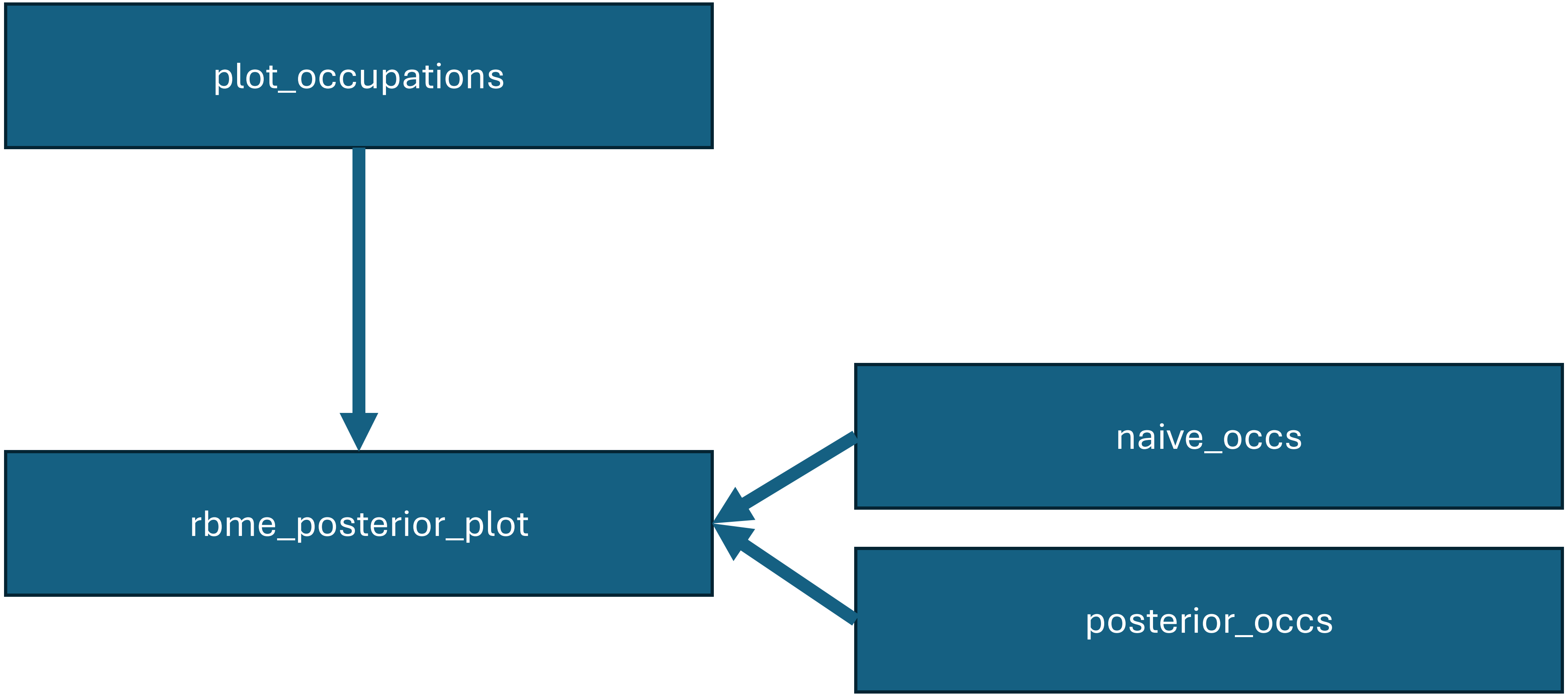
Two parameters: Diffusion coefficient, localization error

2D grid: 100 X 36

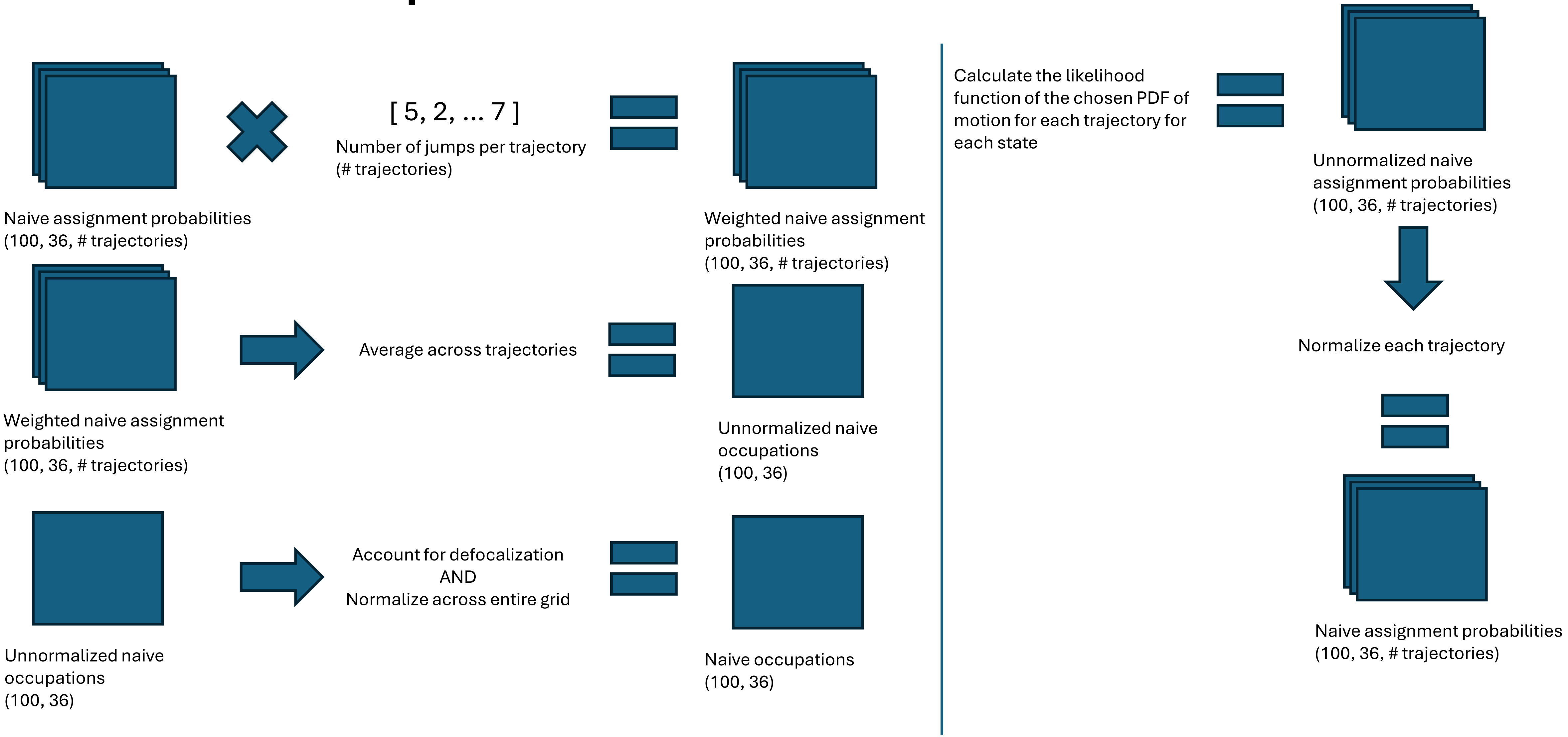
# Inference
SA.plot_occupations(out_file_path)

This function calls a series of other functions that automatically estimates parameters and outputs a graph of results.

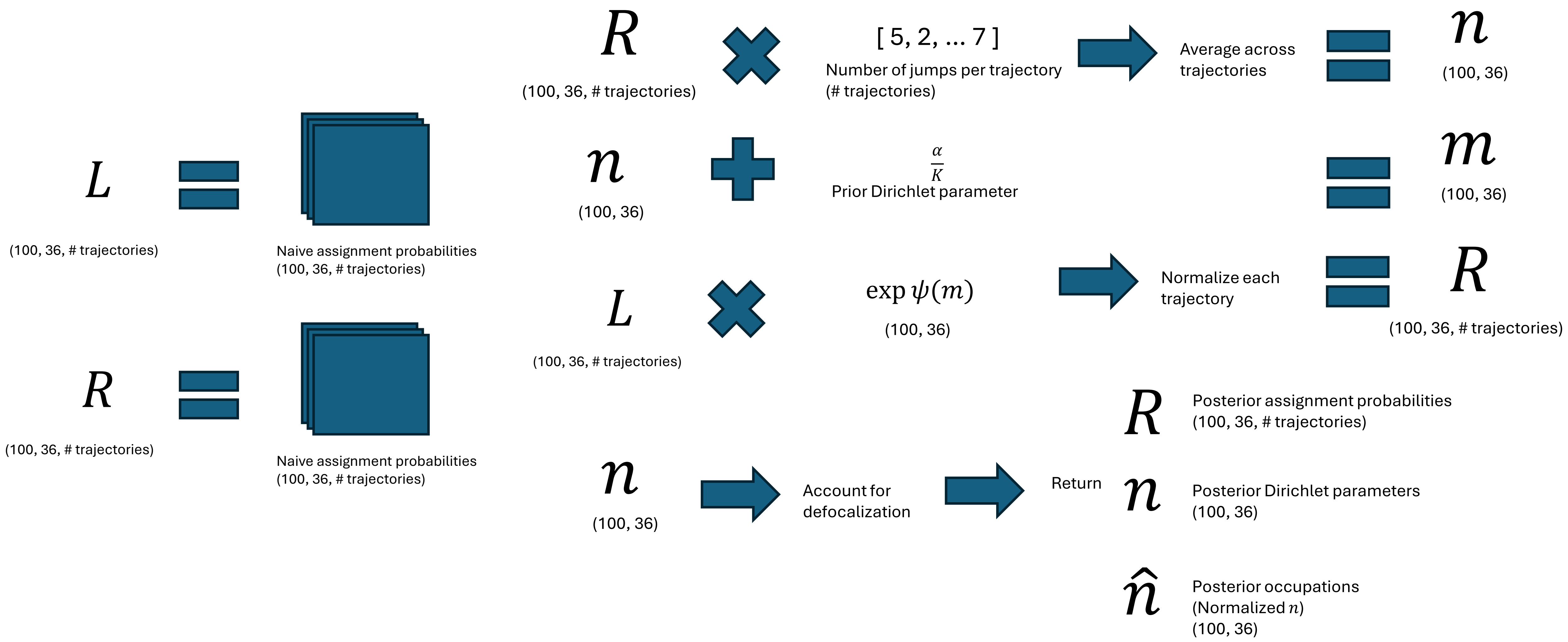


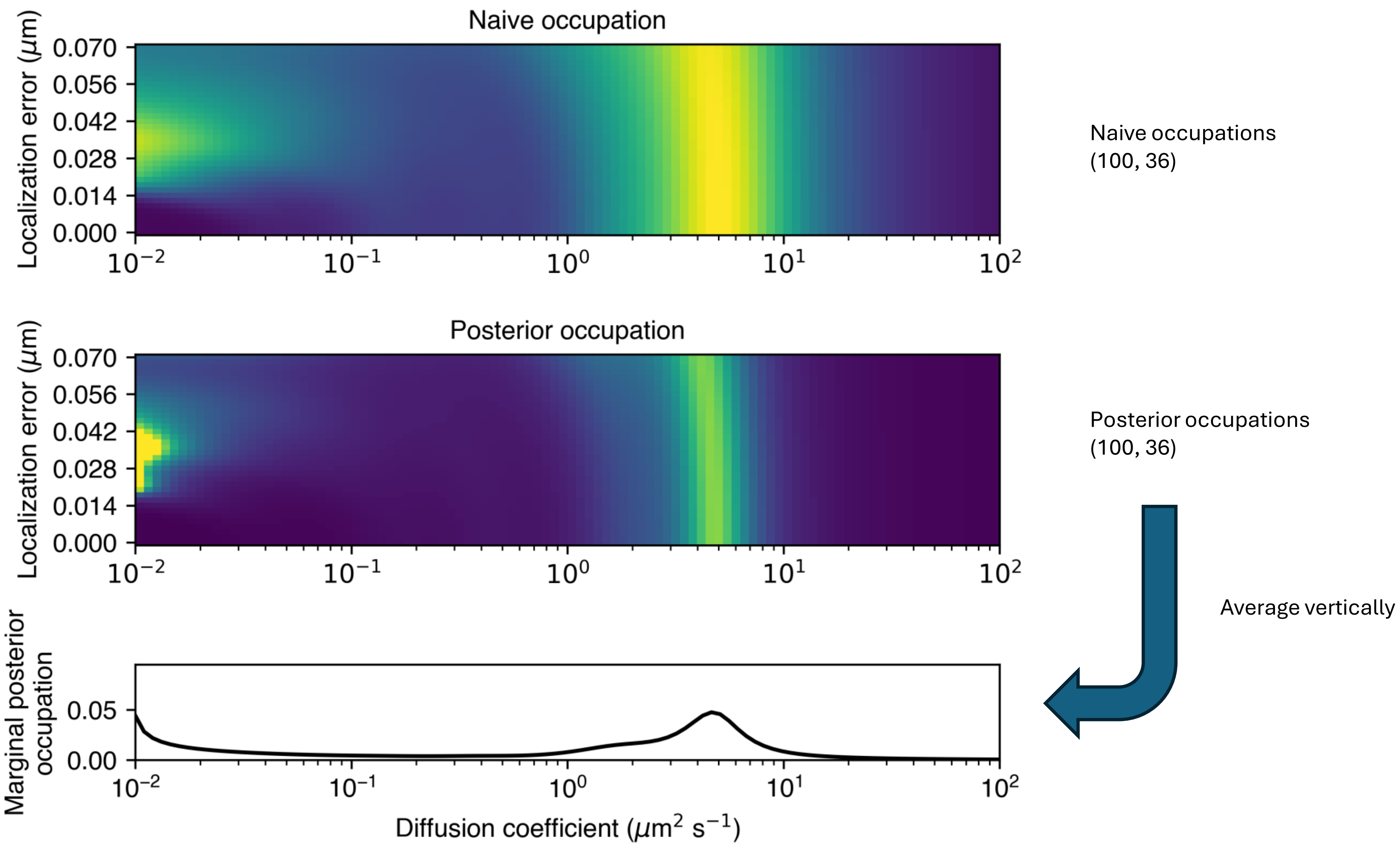

plot_occupations
rbme_posterior_plot
naive_occs
posterior_occs

Naive Occupations



Posterior Occupations




Naive occupation
Localization error (μm)
0.070
0.056
0.042
0.028
0.014
0.000
10^−2
10^−1
10^0
10^1
10^2
Naive occupations
(100, 36)
Posterior occupation
Localization error (μm)
0.070
0.056
0.042
0.028
0.014
0.000
10^−2
10^−1
10^0
10^1
10^2
Posterior occupations
(100, 36)
Average vertically
Marginal posterior occupation
0.05
0.00
10^−2
10^−1
10^0
10^1
10^2
Diffusion coefficient (μm^2 s^−1)